Learning Multimodal Urban Representations for Economic Analogue Retrieval from Satellite Imagery

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1 Background & Motivation

Official economic statistics such as GDP, employment, and building permits are valuable but delayed. BEA metropolitan GDP estimates are typically released months after the observation period, limiting their usefulness for real-time monitoring. Satellite imagery offers a complementary perspective: platforms such as MODIS and VIIRS continuously observe urban expansion, land use, and nighttime activity at global scale and near-zero marginal cost.

Recent advances in representation learning have motivated growing interest in linking satellite imagery with socioeconomic indicators. However, direct economic prediction from imagery remains difficult because urban systems exhibit strong regional variation. Visually similar cities may differ economically due to policy, industrial structure, or regional context. Our EDA confirms that cross-sectional pixel statistics are dominated by city size rather than economic dynamics ($r \approx 0.01$ across the panel), making direct pixel-to-GDP regression ineffective.

Existing satellite-economy work often asks whether imagery can directly predict economic outcomes such as GDP, employment, or poverty. We ask a different question: whether satellite imagery can retrieve economically similar historical city-years after being aligned with economic representations during training. This reframing weakens the assumption that visual urban structure alone determines economic state, while still allowing satellite imagery to support economically meaningful comparison and monitoring.

We therefore frame the task as analogue retrieval rather than direct prediction. Instead of estimating GDP growth directly from imagery, we ask whether satellite-derived urban representations can retrieve economically similar historical city-years in a shared multimodal latent space. Prior work has linked nighttime lights and urban extent to economic activity [1]; we extend this idea by learning multimodal representations that support satellite-only retrieval at inference time.

2 Problem Statement

Given only a city's satellite imagery at time t, we ask whether it is possible to retrieve historical city-years whose economic trajectories serve as useful analogues for the city's current economic state. The primary prediction target is year-over-year real GDP growth rate, measured in percentage points. The unit of observation is a city-year: one annual satellite composite paired with one row of economic indicators. The dataset contains 30 U.S. metropolitan areas from 2013–2023, excluding 2020 due to the COVID-19 structural break. The split is: train 2013–2018 (180 city-years), validation 2019 (30 city-years), and test 2021–2023 (90 city-years).

Table 1: Data sources used in this project.

Source	Variables	Resolution	Coverage
NASA GIBS — MODIS Terra RGB	True-color surface reflectance	250 m	2013–2023
NASA GIBS — VIIRS Day/Night	Nighttime radiance	500 m	2017–2023
BEA CAGDP9	Metro GDP (chained 2017$)	Metro	2013–2023
BLS LAUS	Employment & unemployment	Metro	2013–2023
Census BPS	Building permits issued	Metro	2013–2023
GHSL GHS-BUILT-S R2023A	Built-up surface fraction	$\sim$90 m	2000–2020

3 Data & Exploratory Data Analysis

3.1 Data sources

The dataset combines satellite-derived urban morphology with aligned socioeconomic indicators across 30 U.S. metropolitan areas. MODIS RGB imagery, VIIRS nighttime radiance, and GHSL built-up masks provide complementary views of urban structure, while BEA GDP, BLS employment, and Census building permits provide economic indicators.

MODIS images are 3-band annual composites; VIIRS radiance is spatially aligned to the MODIS grid and globally normalized. GHSL built-up masks are aligned to the same grid during preprocessing. The economic panel merges GDP, employment, unemployment, and growth indicators into a 300-row $\times$ 47-column dataset. Nine economic variables are used as model inputs. Overall missingness is 10.7%, driven primarily by pre-2017 VIIRS unavailability and first-year lag features. Missing values are masked during training rather than imputed.

3.2 Key EDA findings

Satellite signal is weak across cities but stronger within cities over time. Cross-sectional correlations between raw satellite statistics and GDP are nearly zero when all metros are pooled together ($r \approx 0.01$), indicating that simple pixel averages are dominated by fixed city-level differences rather than economic state. Direct satellite-to-GDP regression is therefore unreliable in the cross-sectional setting. Within individual metros, however, temporal satellite variation tracks economic dynamics more meaningfully. VIIRS intensity correlates positively with GDP, employment, and building permits within multiple cities, with correlations commonly ranging from $r = 0.3$ to 0.7. This suggests that satellite imagery contains economically relevant temporal information even though cross-city prediction remains difficult. Pre-pandemic GHSL built-up area growth correlates with GDP growth at $r = 0.52$, substantially stronger than raw pixel statistics. These findings motivate learning city-aware urban representations and multimodal alignment rather than direct pixel-level regression.

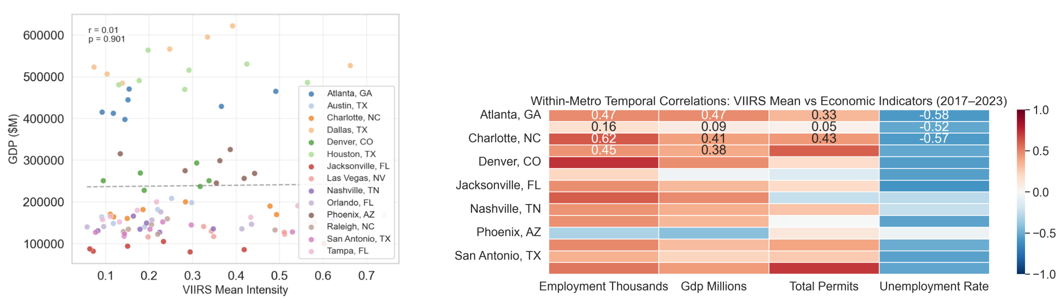

Figure 1: Satellite signal is weak across cities but stronger within cities over time. Left: pooled VIIRS intensity has near-zero correlation with GDP across metros ($r \approx 0.01$). Right: within-metro temporal VIIRS variation more closely tracks GDP, employment, and building permits over time.

COVID-19 introduces a structural break. Economic indicators exhibit a sharp discontinuity in 2020, while built-up urban structure changes much more gradually. This temporal mismatch makes sequence-based forecasting unstable across the pandemic boundary, so 2020 is excluded from all training, validation, and test splits. Before the pandemic, built-up area growth positively correlates with GDP growth ($r = 0.52$), suggesting that urban expansion contains economically meaningful long-term structure.

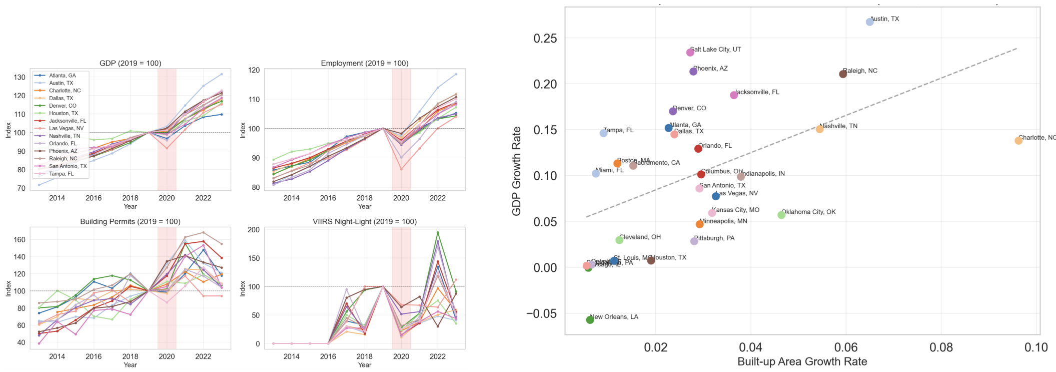

Figure 2: COVID-era structural break and long-term urban expansion signal. Left: GDP, employment, permits, and VIIRS intensity indexed to 2019 baselines show a sharp discontinuity in 2020. Right: pre-pandemic built-up area growth positively correlates with GDP growth across metros ($r = 0.52$).

4 Methods & Models

The framework progressively transforms raw satellite imagery into economically meaningful analogue retrievals through urban representation learning, economic-state compression, multimodal alignment, and satellite-only retrieval.

4.1 GHSL-supervised image encoder

ImageNet-pretrained CNN features capture generic texture and object boundaries but are not optimized for urban morphology relevant to economic activity. We therefore use GHSL built-up masks as auxiliary supervision to teach the encoder urban structure rather than natural-image texture alone. We fine-tune a ResNet-18 backbone [2] pretrained on ImageNet to produce urban-aware 512-dimensional embeddings from 4-channel MODIS+VIIRS input (RGB + night-light). ResNet layers 1–2 are frozen while layers 3–4 are fine-tuned using GHSL supervision. Annual city images are divided into non-overlapping 128×128 patches for training. Optimization uses a 50/50 BCE-Dice loss. Fine-tuning quality is evaluated through centroid separation ratios (between-city distance / within-city distance): the frozen ImageNet encoder achieves 1.2175, while the GHSL-supervised encoder achieves 1.4184 (+16.5%), indicating improved city-level discrimination.

4.2 Economic encoder selection

Raw economic indicators are noisy, partially redundant, and difficult to align directly with image embeddings. We therefore compress multiple economic variables into a lower-dimensional economic-state representation before multimodal alignment. We compare three candidate encoders: an MLP autoencoder operating on individual city-years, and GRU/LSTM autoencoders operating on rolling 3-year sequences. All models compress nine standardized economic indicators into a 32-dimensional latent representation and are trained with Adam, validation-based scheduling, and early stopping. The MLP autoencoder achieves substantially lower reconstruction error than the recurrent alternatives and is selected for the multimodal pipeline, while the GRU and LSTM models generalize poorly across the COVID-era structural break.

4.3 Multimodal VAE with contrastive alignment

To go beyond the methods covered in class, we independently researched recent multimodal contrastive learning approaches, especially KnowCL for urban imagery-based socioeconomic prediction [14] and Google Research's ALIGN blog for general cross-modal embedding alignment [15]. These sources inspired our decision to align satellite embeddings and economic embeddings in a shared latent space.

The next stage aligns satellite representations with economic representations in a shared latent space, allowing satellite imagery to inherit economically meaningful structure during training even though economic inputs are unavailable at inference time. The GHSL-supervised image encoder and selected economic encoder feed into a shared 16-dimensional latent space through a multimodal Variational Autoencoder. Image and economic embeddings are jointly optimized using reconstruction, KL, and InfoNCE contrastive losses:

$$\mathcal{L} = \mathrm{MSE}_{\mathrm{img}} + \mathrm{MSE}_{\mathrm{econ}} + \beta D_{\mathrm{KL}} + \lambda_c \mathcal{L}_{\mathrm{InfoNCE}}$$

KL warmup is used to reduce posterior collapse. The InfoNCE loss aligns paired image/economic observations while separating unrelated city-years, encouraging economically similar cities to occupy nearby regions of latent space. During training, one modality is randomly zeroed with total probability 0.3: image embeddings are dropped with probability 0.15 and economic embeddings with probability 0.15.

4.4 Satellite-only analogue retrieval

Direct GDP prediction from imagery assumes that visual urban structure uniquely determines economic state, an assumption that is too strong under geographic heterogeneity. Instead of directly predicting GDP growth, the final stage retrieves historical analogues from latent space to smooth uncertainty through neighborhood averaging. At inference time, all economic inputs are removed: MODIS+VIIRS images are encoded into the shared latent space using the image branch alone, and nearest neighbors are retrieved from the training latent space. Retrieved neighbors' GDP-growth values are averaged to produce the final estimate. A validation search on 2019 GDP growth compares cosine, scaled Euclidean, and scaled Manhattan retrieval rules across multiple neighborhood sizes, and the best validation rule is frozen before final test evaluation.

5 Results & Interpretation

5.1 GHSL supervision improves urban representations

Direct GHSL segmentation transfer remains difficult at the 30-city scale, with mean validation IoU $= 0.0037$ and Dice $= 0.0071$. The low scores mainly reflect the resolution mismatch between MODIS (250 m) and GHSL ($\sim$90 m), indicating that fine urban boundaries cannot be fully recovered from coarse imagery. However, the segmentation task primarily serves as auxiliary supervision rather than the final objective. The key result is that GHSL supervision improves representation quality: centroid separation increases from 1.2175 for the frozen ImageNet encoder to 1.4184 after GHSL fine-tuning (+16.5%), indicating that the encoder learns more city-discriminative urban structure rather than generic ImageNet texture.

5.2 Economic-state compression favors non-sequential representations

Table 2: Economic autoencoder comparison on the 2021–2023 holdout. Lower MSE is better.

Model	Test MSE
MLP autoencoder	**0.0559**
GRU autoencoder	1.0810
LSTM autoencoder	1.1920

The MLP autoencoder substantially outperforms the recurrent alternatives. The GRU and LSTM models generalize poorly across the COVID-era structural break, suggesting that compact non-sequential economic-state representations are more stable under temporal distribution shift.

5.3 Multimodal alignment organizes economic structure in latent space

The joint multimodal decoder achieves a validation MSE of 0.4895 on economic reconstruction, outperforming the per-city training-mean baseline (0.5548). However, image-only decoding remains

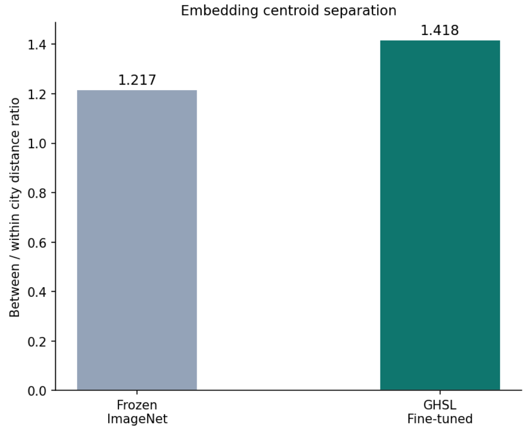

Figure 3: Centroid separation ratio for frozen ImageNet and GHSL fine-tuned embeddings. GHSL supervision improves city-level separation by 16.5%.

weaker (0.8156), indicating that direct economic prediction from imagery is unreliable in this setting. Instead, the main contribution of the multimodal alignment stage is latent-space organization. A UMAP projection of the shared latent space shows visible economic structure: regions with higher GDP growth separate from regions with higher unemployment, suggesting that the alignment process transfers economically meaningful geometry into the image latent space.

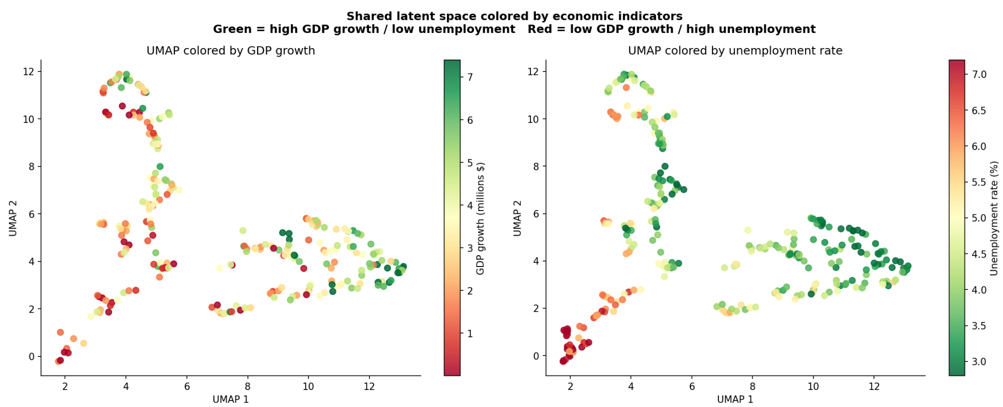

Figure 4: UMAP projection of the shared latent space colored by GDP growth rate and unemployment rate.

5.4 Satellite-only analogue retrieval outperforms naive baselines

Table 3: GDP-growth-rate retrieval benchmark on the 2021–2023 holdout. Lower MAE in percentage points is better.

Method	Test MAE
Random retrieval (avg. 100 draws)	2.923
Training-set mean	2.558
Best plain-cosine retrieval	2.504
Scaled Euclidean k=8	**2.435**
Previous-year economic value	2.358

The selected retrieval rule outperforms all naive baselines and remains within 0.078 percentage-point MAE of the previous-year economic baseline despite using no economic inputs at inference time. This result supports the central framing of the project: direct GDP prediction from imagery is too strong an assumption, but economically meaningful analogue retrieval remains feasible after multimodal alignment. Expanding the dataset from earlier experiments to 30 metros substantially increases geographic and socioeconomic heterogeneity, making retrieval more difficult while providing a more realistic robustness test. Scaled Euclidean retrieval outperforms plain cosine retrieval, suggesting that variance-normalized latent distances better capture economic similarity. Retrieval errors are largest for rapidly growing metros whose post-COVID trajectories differ substantially from historical analogues in the training set.

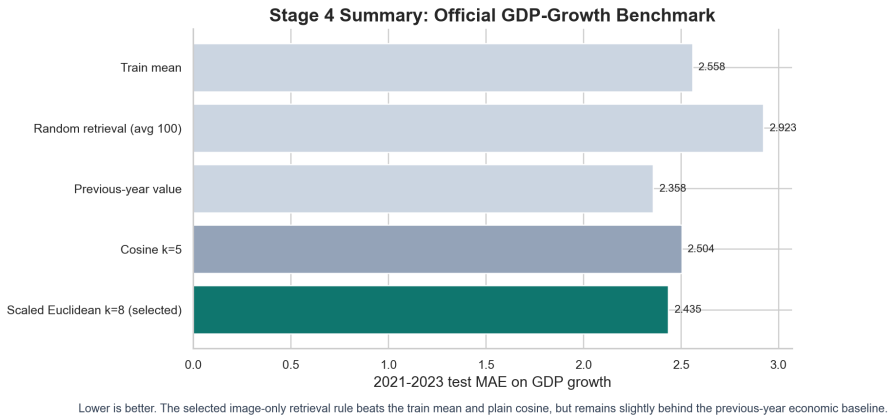

Figure 5: GDP-growth-rate retrieval benchmark on the 2021–2023 holdout.

5.5 Illustrative analogue retrieval patterns

Table 4: Illustrative examples of the types of analogue relationships the retrieval framework is designed to capture.

Query city-year	Example analogue city-years	Expected GDP-growth pattern	Why this is meaningful
Austin 2022	Raleigh 2018; Denver 2017; Nashville 2019	Higher-growth expanding metros	Illustrates retrieval of rapidly growing metros with similar urban expansion dynamics.
Detroit 2021	Cleveland 2018; Pittsburgh 2017; St. Louis 2019	Lower-growth legacy metros	Illustrates retrieval of historically industrial metros with slower post-industrial growth trajectories.

Compared with the earlier MS3 pipeline, the final framework removes all economic inputs at inference time and replaces hand-engineered satellite summaries with learned multimodal representations. The final system therefore evaluates a harder but more realistic satellite-only retrieval setting.

6 Conclusions & Discussion

What we can claim. On a 30-metro panel spanning 2013–2023, the proposed satellite-only retrieval framework achieves economically meaningful analogue retrieval without economic inputs at inference time. The system outperforms the training-set mean, random retrieval, and plain-cosine baselines through the combined effect of GHSL-supervised image encoding, multimodal latent alignment, and tuned retrieval. More broadly, the project demonstrates a reusable framework for learning satellite-based urban representations that support economic analogue discovery when traditional economic data are delayed, sparse, or unavailable.

What we cannot claim. The system is not a general-purpose economic forecaster. Direct image-only decoding of economic state is weaker than a simple per-city mean. We cannot claim that the approach generalizes beyond the 30-city panel or to economic indicators not evaluated here.

Limitations. Several limitations remain. First, the dataset contains only 180 training city-years, which is relatively small for multimodal deep learning and may limit representation robustness. Second, the COVID-era structural break introduces substantial temporal distribution shift between training and evaluation periods. Third, MODIS imagery at 250 m resolution cannot capture fine urban structure visible in GHSL at 90 m resolution, limiting the precision of morphology supervision. Finally, the evaluation focuses specifically on GDP growth, so the results may not generalize to other economic indicators without additional validation.

Future work. Extending the metro panel, incorporating higher-resolution imagery (e.g., Sentinel-2), and testing leave-one-metro-out evaluation would improve geographic generalization. Future systems could also replace point estimates with uncertainty-aware analogue distributions.

7 Broader Impact

Satellite-based representation learning may improve urban analysis in regions where traditional economic statistics are sparse or delayed. Such systems could support infrastructure planning, development monitoring, and comparative policy analysis at larger geographic scale.

However, urban economic conditions cannot be fully inferred from visual morphology alone. Models trained on geographically uneven datasets may encode regional biases and perform poorly in under-represented areas. More broadly, this work highlights both the promise and limitations of multimodal machine learning for urban analysis under geographic diversity.

References

[1] Henderson, J. V., Storeygard, A., & Weil, D. N. (2012). Measuring economic growth from outer space. *American Economic Review*, 102(2), 994–1028. doi:10.1257/aer.102.2.994.

[2] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778. doi:10.1109/CVPR.2016.90.

[3] Pesaresi, M., & Politis, P. (2023). GHS-BUILT-S R2023A — GHS built-up surface grid, derived from Sentinel-2 composite and Landsat, multitemporal (1975–2030). European Commission, Joint Research Centre (JRC) [Dataset]. doi:10.2905/9F06F36F-4B11-47EC-ABB0-4F8B7B1D72EA.

[4] NASA Global Imagery Browse Services (GIBS). MODIS Terra Corrected Reflectance (True Color) imagery and VIIRS Day/Night Band nighttime imagery. NASA Earthdata / EOSDIS. Accessed May 2026.

[5] U.S. Bureau of Economic Analysis. CAGDP9: Real GDP by county and metropolitan area. BEA Regional Data [Dataset]. Accessed May 2026.

[6] U.S. Bureau of Labor Statistics. Local Area Unemployment Statistics (LAUS): metropolitan area employment and unemployment data. BLS [Dataset]. Accessed May 2026.

[7] U.S. Census Bureau. Building Permits Survey (BPS). U.S. Census Bureau [Dataset]. Accessed May 2026.

[8] Kingma, D. P., & Welling, M. (2014). Auto-Encoding Variational Bayes. *International Conference on Learning Representations (ICLR)*. arXiv preprint arXiv:1312.6114.

[9] van den Oord, A., Li, Y., & Vinyals, O. (2018). Representation learning with contrastive predictive coding. arXiv preprint arXiv:1807.03748.

[10] McInnes, L., Healy, J., Saul, N., & Großberger, L. (2018). UMAP: Uniform Manifold Approximation and Projection. *Journal of Open Source Software*, 3(29), 861. doi:10.21105/joss.00861.

[11] Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., Desmaison, A., Köpf, A., Yang, E., DeVito, Z., Raison, M., Tejani, A., Chilamkurthy, S., Steiner, B., Fang, L., … Chintala, S. (2019). PyTorch: An imperative style, high-performance deep learning library. *Advances in Neural Information Processing Systems*, 32.

[12] TorchVision Contributors. (2026). ResNet-18 model documentation and pretrained ImageNet weights. PyTorch / TorchVision Documentation. `https://pytorch.org/vision/`

[13] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., VanderPlas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

[14] Liu, Y., Zhang, X., Ding, J., Xi, Y., & Li, Y. (2023). Knowledge-infused contrastive learning for urban imagery-based socioeconomic prediction. *Proceedings of the ACM Web Conference 2023 (WWW '23)*, 4150–4160. doi:10.1145/3543507.3583876.

[15] Google Research. (2021). ALIGN: Scaling Up Visual and Vision-Language Representation Learning With Noisy Text Supervision. Google Research Blog, May 11, 2021.

[16] Generative AI assistance: OpenAI ChatGPT and Anthropic Claude were used for debugging assistance, LaTeX formatting, slide outlining, and writing refinement during project development and report preparation. All final analyses, experiments, code, and interpretations were reviewed and validated by the authors.

A Metro List

Table 5: The 30 U.S. metropolitan areas used in this study.

Atlanta, GA	Detroit, MI	Miami, FL	Pittsburgh, PA
Austin, TX	Houston, TX	Minneapolis, MN	Raleigh, NC
Boston, MA	Indianapolis, IN	Nashville, TN	Sacramento, CA
Charlotte, NC	Kansas City, MO	New Orleans, LA	Salt Lake City, UT
Chicago, IL	Las Vegas, NV	Orlando, FL	San Antonio, TX
Cleveland, OH	Los Angeles, CA	Philadelphia, PA	St. Louis, MO
Columbus, OH	Denver, CO	Phoenix, AZ	Tampa, FL
Dallas, TX	Washington, DC		

B Retrieval Rule Tuning Results

Table 6: Validation-search results for retrieval rule selection.

Model	Metric	k	Val MAE
Scaled Euclidean k=8	euclidean	8	**1.159**
Scaled Euclidean k=5	euclidean	5	1.203
Scaled Manhattan k=8	manhattan	8	1.214
Scaled Manhattan k=5	manhattan	5	1.231
Scaled Euclidean k=3	euclidean	3	1.287
Cosine k=3	cosine	3	1.318
Cosine k=5	cosine	5	1.341
Cosine k=2	cosine	2	1.392
Cosine k=1	cosine	1	1.504

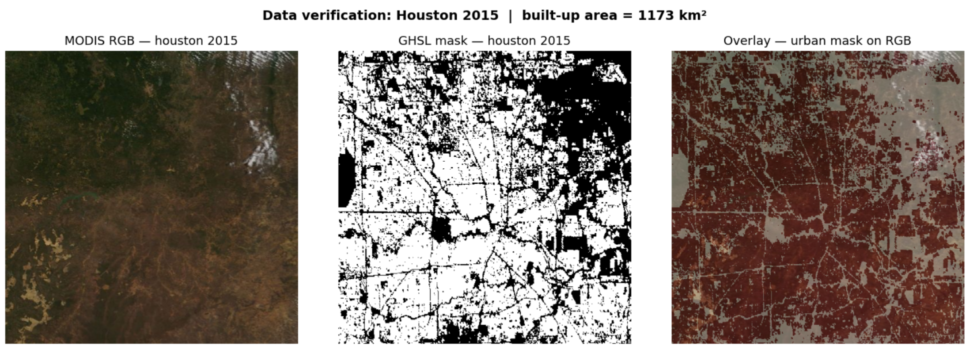

Figure 6: MODIS RGB imagery, GHSL built-up mask, and overlay alignment for Houston (2015), used to verify spatial consistency between satellite imagery and GHSL supervision.